

DBMS MINI PROJECT

FoodFlow

Online Food Delivery Platform

A user-friendly digital platform connecting customers, restaurants and delivery operations.

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1. Project Overview

What the platform solves and who benefits

PROJECT CLARITY



Customer



Browse restaurants and menus, add items to cart, place orders, make payments, track order status and share orders with a group.



Restaurant



Manage menu items, receive customer orders, update preparation status and monitor order activity.



Delivery & Admin



Coordinate delivery, monitor platform activity, support users and view operational information.

Core idea: one platform → one connected ordering journey → better convenience and operational visibility.

2. Problem Statement & Societal Need

Why an integrated food delivery platform is relevant

Customers often switch between restaurant discovery, ordering, payment and delivery-status channels.

Restaurants need a simple digital way to manage menus and incoming orders.

Group meals can create confusion when several people order together and need to split the bill.

Customers expect faster status updates and transparent order progress.

Local restaurants benefit from a digital channel that can extend their reach beyond walk-in customers.

SOCIAL IMPACT

Convenience

Order from anywhere

Accessibility

Digital access to local food providers

Efficiency

Less manual order handling

Transparency

Clear order-status visibility

Collaboration

Shared group ordering and bill splitting

3. Objectives

Clear, measurable goals for the proposed system

01 Simplify food discovery

Provide restaurant and menu browsing through a single interface.

02 Streamline ordering

Support cart management, checkout and order placement.

03 Improve visibility

Show order progress and delivery-related updates.

04 Support restaurants

Give restaurants tools to manage menu and orders.

05 Enable group ordering

Allow shared carts, item ownership and split billing.

06 Maintain secure access

Use role-based authentication and protected user actions.

4. Project Feasibility Analysis

A practical project aligned with the evaluation rubric

Technical Feasibility

React/Vite frontend, Node.js/Express backend and MySQL provide a familiar full-stack architecture. Socket.IO supports real-time updates.

Economic Feasibility

The system can be developed with widely available open-source technologies and standard development tools.

Operational Feasibility

Role-based dashboards and simple workflows make the system understandable for customers, restaurants and administrators.

Schedule Feasibility

The platform is modular: authentication, restaurants, cart, orders, payments, delivery and group ordering can be developed incrementally.

Feasibility takeaway: the proposed solution is technically achievable, affordable for an academic prototype and useful in a real-world context.

5. Existing System vs Proposed System

How FoodFlow improves the conventional ordering experience

Aspect	Conventional / fragmented	FoodFlow proposed platform
Ordering	Multiple steps/channels	Integrated browse → cart → checkout → order
Restaurant management	Manual or limited digital control	Digital menu and order management
Order visibility	Status may be unclear	Status updates across the order lifecycle
Group meals	Separate orders and manual bill calculation	Shared cart + item ownership + split bill
Communication	Static confirmation	Real-time Socket.IO updates
Scalability	Difficult to extend	Modular architecture for future services

6. Novelty of the Project

The feature that differentiates FoodFlow

NOVELTY FEATURE

1

Create group

Generate a shared group order.

2

Shared cart

Each member adds and owns their items.

3

Split bill

Calculate individual contributions.



Group Ordering with Split Bill

FoodFlow introduces a collaborative ordering experience for friends, classmates, families or office teams. A group can use one shared cart while every member retains item ownership. The platform calculates the group total and supports individual bill splitting and payment handling, while Socket.IO can broadcast live updates to group members.

Shared Cart

One group order

Ownership

Member-wise items

Split Bill

Individual contribution

Live Updates

Socket.IO events

✓ **Split Bill Feature: the final group bill is divided among participants for individual payment.**

7. Technology Stack & Architecture

Full-stack implementation used for the FoodFlow prototype

Frontend

React + Vite — Responsive user interface

Backend

Node.js + Express — REST APIs and business logic

Database

MySQL — Users, restaurants, menu, orders and payments

Real-time

Socket.IO — Live order/group updates

Security

JWT + bcryptjs — Authentication and password protection

Integration

Razorpay + Maps — Payments and location services



8. Major Modules

Functional decomposition of the platform

1 Customer Module

Registration, login, restaurant browsing, menu, cart, checkout, orders

2 Restaurant Module

Restaurant profile, menu management, order acceptance and status

3 Delivery Module

Delivery assignment, status updates and location-related services

4 Admin Module

User/restaurant/order monitoring and platform administration

5 Payment Module

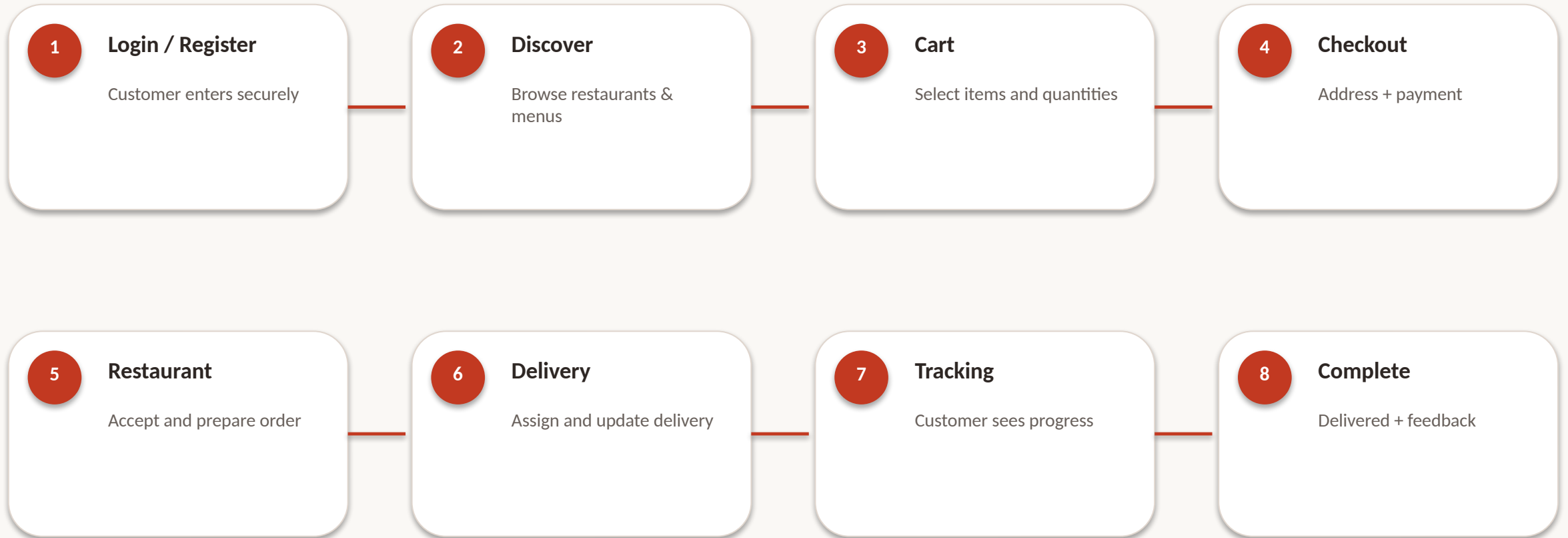
Checkout, payment processing and transaction status

6 Group Ordering

Shared group code, shared cart, member ownership and split bill

9. End-to-End System Workflow

From discovery to delivery



10. Database Design & DBMS Relevance

Core entities for reliable order management

CUSTOMER

customer_id
name • email • address

RESTAURANT

restaurant_id
name • location

MENU_ITEM

item_id
name • price • category

ORDER

order_id
customer • total • status

ORDER_ITEM

order_item_id
order • item • quantity

PAYMENT

payment_id
order • amount • status

DELIVERY

delivery_id
order • partner • status

GROUP_ORDER

group_id
members • shared cart •
split bill

DBMS focus: structured storage, relationships, constraints, transaction records and efficient retrieval of operational data.

11. Expected Results & Benefits

What the project delivers

For Customers

Faster order

For Restaurants

Digital men

For Delivery Operations

Clear order

For Administrators

Centralized

For Society

Supports d

Outcome: a connected, practical and scalable food-delivery prototype.

12. Future Scope

How FoodFlow can evolve beyond the academic prototype

AI-based restaurant and food recommendations
Advanced delivery-route optimization and ETA prediction
GPS-based live delivery tracking with map integration
Multilingual and accessibility-focused interfaces
Loyalty points, offers and subscription plans
Analytics dashboard for restaurant sales and customer trends
Cloud deployment and scalable microservice architecture
Mobile application for customers, restaurants and delivery partners

NEXT-GEN PLATFORM



NOW

Core ordering



NEXT

Tracking + analytics



LATER

AI + optimization



SCALE

Cloud + mobile

13. Rubric Alignment — Targeting 50/50

Directly addressing the five evaluation dimensions shown in the provided rubric

Clarity of Project Title	FoodFlow – Online Food Delivery Platform	Specific, understandable and directly describes the system.
Project Feasibility Analysis	Technical + Economic + Operational + Schedule	Shows that the project is achievable with available tools and resources.
Oriented with Societal Needs	Convenience + local business support + group ordering	Addresses real-world digital ordering and collaboration needs.
Novelty of the Project Title	Group Ordering + Split Bill	Distinctive feature gives the project a clear differentiating point.
Presentation Skills	Structured flow + visuals + live demo	Use the slide sequence to explain problem → solution → novelty → results.

TOTAL RUBRIC: 50 MARKS

Thank You

FoodFlow — Online Food Delivery Platform

A practical DBMS project designed around convenience, connectivity and collaborative ordering.

Reference basis: online food-delivery architecture and project-presentation structures reviewed from public technical/project sources.
Project-specific content reflects the FoodFlow implementation discussed for this DBMS project.